

SPECIFICATION

Ground Floor Lighting and Ceiling Fans

Proposed Residential Building For Mr Chandrasekhar

Anantapur, Andhra Pradesh

Document	GF/EL/SPEC
Revision	A — 22.08.2026
Status	For construction
Accompanying drawings	GF/EL/01 Lighting and ceiling fan layout
	GF/EL/02 Luminaire and fan schedules
	GF/EL/03 Illumination calculations
Read with	ECS __/AR/R2 • STRUCTURAL WD-1

1 SCOPE

- 1.1 This specification covers the supply and installation of general lighting and ceiling fans on the ground floor, together with the conduit, junction boxes and fan hooks that must be cast into the ground floor roof slab.
- 1.2 Excluded from this scope: task lighting of any kind, being under counter, wardrobe, bathroom mirror and desk lighting; switching and switch boards; socket outlets; distribution boards, cabling and earthing; air conditioning; exhaust ventilation. These are covered by separate drawings.
- 1.3 Staircase lighting is excluded. The stairwell is open to the first floor and its luminaires are carried on the first or second floor slab.

2 STANDARDS

- 2.1 IS 3646 (Part 1) : 1992 Code of practice for interior illumination — principles of good lighting and aspects of design.
- 2.2 National Building Code of India 2016, Part 8 Section 1, Lighting and Ventilation, and Part 8 Section 2, Electrical and Allied Installations.
- 2.3 IS 732 : 2019 Code of practice for electrical wiring installations.
- 2.4 IS 9537 (Part 3) Conduits for electrical installations, rigid plain conduits of insulating material.
- 2.5 IS 10322 (Part 5) Luminaires, particular requirements. LED luminaires to IS 16107 and IS 16108 as applicable.
- 2.6 IS 374 Electric ceiling type fans and regulators. BLDC fans to carry a valid BEE star label.

3 LUMINAIRES

- 3.1 All luminaires are LED surface panels with integral driver, mounted directly on the RCC soffit. There is no false ceiling anywhere on this floor and no recessed fitting is to be used.

- 3.2** Minimum luminous efficacy 100 lm/W at the luminaire. Minimum colour rendering index 80 generally and 90 in the kitchen and bathrooms. Colour consistency within 3 SDCM across all fittings of one type.
- 3.3** Rated life not less than L70B50 at 25000 hours. Driver power factor not less than 0.90 at full load. Surge protection not less than 2 kV line to neutral.
- 3.4** Wet locations: bathrooms and utility to IP44 minimum, portico to IP54 minimum with a UV stabilised diffuser.
- 3.5** Luminaire types, quantities and locations are scheduled on drawing GF/EL/02. Any proposed equivalent is to be submitted with photometric data and approved before ordering.

TYPE	DESCRIPTION	RATING	OUTPUT	CCT	IP	QTY
A	LED surface panel luminaire	15 W	1500 lm	3000 K	IP20	9
B	LED surface panel luminaire	18 W	1800 lm	3000 K	IP20	6
C	LED surface panel luminaire	22 W	2200 lm	4000 K	IP20	4
D	LED surface panel luminaire	18 W	1800 lm	4000 K	IP20	4
E	LED surface panel luminaire	15 W	1500 lm	4000 K	IP44	5
F	LED surface panel luminaire	24 W	2400 lm	3000 K	IP54	2

4 CEILING FANS

- 4.1** All fans are 1200 mm, being 48 in, sweep with a brushless direct current motor, rated not more than 32 W at full speed and delivering not less than 210 CMM.
- 4.2** Fans are supplied with an integral radio frequency receiver in the canopy and a handheld remote. The wall position is therefore a plain switched live; no step type regulator is to be fitted, as it will not control a BLDC fan and may damage the electronics.
- 4.3** Downrod length 600 mm, being 24 in, in all internal rooms and 900 mm, being 36 in, in the portico, so that the blade plane is held at 8'-3½" above finished floor internally. This is not an optional item: with a 10'-6½" soffit the stock 250 mm rod leaves the blades too close to the ceiling to move air effectively.
- 4.4** Minimum blade clearance to any wall 12 in. Minimum blade level above finished floor 2100 mm in accordance with normal practice.
- 4.5** The portico fan is to be an outdoor rated model with a powder coated or anodised finish and sealed motor housing, suitable for a dusty inland location.

5 WORK TO BE CAST INTO THE SLAB

- 5.1** Fan hooks. A closed loop bent from 10 mm diameter mild steel bar, hooked over two bottom mesh bars in each direction and tied with binding wire at all four crossings. The loop is to project 2½ in to 3 in below the finished soffit and to be capable of carrying a static load of 100 kg. Hooks tied to a single bar are not acceptable.
- 5.2** Conduit. ISI marked medium gauge rigid PVC to IS 9537 (Part 3). 20 mm diameter for up to three 1.5 sq mm cores and 25 mm diameter above that. Conduit is to be laid between the top and bottom layers of slab reinforcement, never above the top mesh, with not less than 25 mm cover to the soffit. A galvanised iron draw wire is to be left in every run.
- 5.3** Beams. No conduit is to pass through any reinforced concrete beam. Runs are to be routed around the beam within the slab panel, or dropped down the wall and raised again on the far side. The layout on GF/EL/01 is arranged so that no run needs to cross a beam.
- 5.4** Junction boxes. 75 mm diameter PVC boxes set flush with the finished soffit. Not more than two 90 degree bends between draw points; where a third bend is unavoidable an additional box is to be provided. No box is to be concealed behind finishes.
- 5.5** Sunken zones. Over the Common Bath, the Handwash, the south strip of Bedroom 02 and the Master Bath the slab panel steps down by approximately 12 in. Conduit is to follow the step and is not to bridge across it.

6 SEQUENCE AND INSPECTION

- 6.1 The electrical contractor is to work on the deck after the bar bender has completed the bottom mesh and before the top mesh is laid.
- 6.2 Before any concrete is placed, the contractor is to walk the deck with the client and the architect and obtain confirmation of every fan hook and light point position, and is then to photograph the entire deck with a tape rule in frame.
- 6.3 All draw wires are to be pulled through again within one week of the pour and before the shuttering is struck. Any run found to be blocked is to be reported immediately.
- 6.4 Insulation resistance is to be tested with a 500 V megger after wiring and before energising, and a test record submitted.

7 ASSUMPTIONS TO BE CONFIRMED

- 7.1 The depth of the first floor sunken slab is taken as 12 in. It governs the soffit level in Bedroom 02, the Common Bath, the Handwash and the Master Bath, and therefore the mounting level of six luminaires. To be confirmed by the structural engineer before setting out.
- 7.2 Beam depths where the structural drawing carries no type mark have been inferred. To be confirmed against the marked up structural print.
- 7.3 Room dimensions have been taken from the architect's drawing ECS ___/AR/R2 and reproduce to within 15 mm. Any dimensional change is to be notified before the fittings are ordered.